

Heat and Temperature

Subject: Physics

Grade: Grade 9

Explore how heat moves, how temperature is measured, and why materials expand when they warm.

Heat vs Temperature

Temperature measures how hot something is. Heat is the energy transferred because of temperature difference.

Methods of Heat Transfer

Heat moves by conduction, convection, and radiation. Each method works differently in solids, liquids, and gases.

Thermal Expansion

Materials expand when heated. This is why bridges and rails need gaps so they can grow without bending.

Insulation and Everyday Uses

Insulators slow heat flow, while conductors transfer heat quickly. This is why houses use insulation and cooking pots use metal.